Clustering Analysis

Detailed Technical Report with Figures

Online Retail II · Principles of Data Science 7144 · Group Assignment

Dataset	Online Retail II (Kaggle) — 333,234 cleaned records
Customers	4,191 raw → 4,124 after outlier removal (67 removed)
Products	3,575 unique products clustered into demand tiers
Countries	37 countries grouped into 3 market performance tiers
Methods	K-Means + Hierarchical Clustering (Ward linkage) + PCA Visualisation
Validation	Adjusted Rand Index: Customer ARI = 0.762 · Country ARI = 0.795
Figures	16 figures embedded directly from 03_clustering.ipynb

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Part 1 — Customer Segmentation (RFM Clustering)

Retail transaction data is well-suited for customer segmentation using RFM analysis. By computing three metrics per customer — Recency, Frequency, and Monetary value — a rich feature space is created that captures each customer's engagement level. Both K-Means and Hierarchical Clustering are applied and cross-validated using the Adjusted Rand Index.

RFM Feature Definitions

Feature	Description	Formula
Recency (R)	Days since customer's last purchase	Reference Date – Last Invoice Date
Frequency (F)	Number of unique invoices placed	Count of unique Invoice IDs per customer
Monetary (M)	Total revenue generated	Sum of Total_Revenue per customer

9-Step Clustering Workflow

Step 1 — Data Quality Check

Confirmed zero missing values, zero duplicates, zero negative quantities or prices in the cleaned dataset.

Step 2 — Feature Engineering

RFM computed per Customer ID using positive-quantity transactions only (returns excluded). Reference date = 2011-12-09.

Step 3 — Outlier Removal

Z-score threshold of 3 applied across all RFM dimensions. 67 of 4,191 customers removed, leaving 4,124 for analysis.

Step 4 — Feature Scaling

StandardScaler applied: all features scaled to mean ≈ 0, std ≈ 1.

Step 5 — Optimal k Selection

Elbow Method (WCSS) and Silhouette Score evaluated for k = 1–10. Silhouette peaks at k = 2 (0.518); k = 4 chosen for business granularity.

Step 6 — K-Means Clustering

KMeans fitted: k=4, random_state=42, n_init=10.

Step 7 — PCA Visualisation

PCA reduces 3D RFM to 2D for scatter plot visualisation of cluster structure.

Step 8 — Cluster Profiling

Mean RFM per cluster computed; each cluster assigned a business-meaningful label.

Step 9 — HC Validation

AgglomerativeClustering (Ward, k=4) applied independently; ARI measured against K-Means.

Data Quality Check: Missing values = 0 · Duplicates = 0 · Negative quantities = 0 · Negative prices = 0 · Dataset: 333,234 records · 4,191 customers identified.

Outlier Removal Detail

Stage	Count	Notes
Raw RFM dataset	4,191	All customers with ≥1 positive-quantity transaction
After z-score filter (threshold = 3)	4,124	67 extreme-value customers removed
Example outlier: Customer 12471	Freq = 26, Mon = £8,703	Extreme values would distort cluster centroids

Optimal k Selection — Elbow Method & Silhouette Score

Both methods are evaluated across k = 1–10 on the standardised RFM data. The WCSS elbow shows diminishing returns beyond k = 4, while the silhouette score peaks at k = 2. k = 4 is chosen to yield four actionable CRM segments consistent with standard retail segmentation practice.

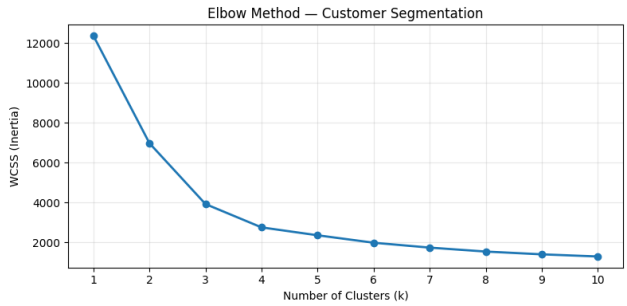

Figure 1 — Elbow Method (WCSS). Inflection around k = 4 supports the cluster count selection.

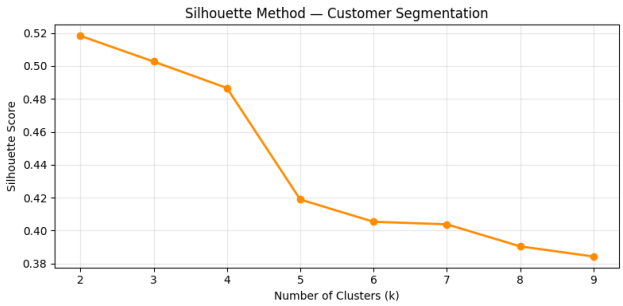

Figure 2 — Silhouette Score. Peaks at k = 2 (score: 0.518). k = 4 chosen for business interpretability.

K-Means Cluster Results (k = 4)

Cluster	Label	Recency	Frequency	Monetary	Count	% Share
2	Loyal High-Value	12.7 days	14.5	£4,158	208	5.0%
3	Active Regular	31.5 days	6.6	£1,813	798	19.4%
0	Occasional Buyers	49.9 days	2.2	£457	2,126	51.6%
1	Lapsed / At-Risk	250.8 days	1.4	£285	992	24.1%

PCA Cluster Visualisation (K-Means, k = 4)

PCA reduces the standardised 3D RFM space to 2 principal components for visualisation. The scatter confirms clear visual separation between all four segments, particularly the Loyal High-Value cluster which is isolated at low recency and high frequency/monetary.

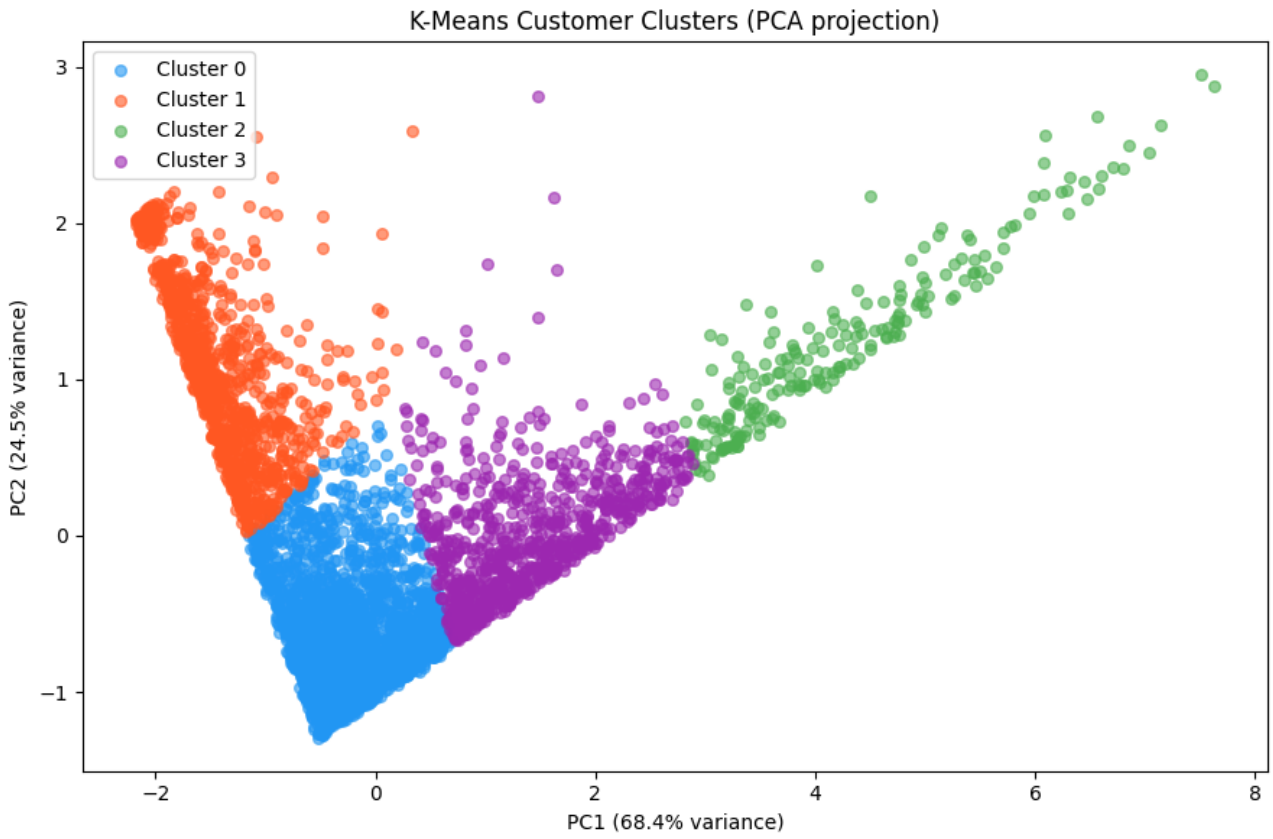

Figure 3 — K-Means Customer Clusters (PCA Projection, k=4). Four distinct groups are visible. Loyal High-Value (top-left, low recency) and Lapsed/At-Risk (bottom-right, high recency) are furthest apart.

Business Interpretation

Cluster	Label	RFM Profile	CRM Strategy
2	Loyal High-Value	Low R, High F&M;	VIP loyalty programmes, exclusive early access, reward schemes
3	Active Regular	Low-Mid R, Mid F&M;	Upsell and cross-sell campaigns to migrate into high-value tier
0	Occasional Buyers	Mid R, Low F&M;	Seasonal promotions and re-engagement emails to increase frequency
1	Lapsed / At-Risk	Very High R, Low F&M;	Win-back campaigns with discount incentives; deprioritise if budget limited

Part 1b — Hierarchical Clustering Validation (Ward, k = 4)

Hierarchical clustering builds a tree of merges without requiring a pre-specified k. Ward linkage is used — minimising within-cluster variance at each step, the same objective as K-Means, making comparison meaningful. The dendrogram is cut at k = 4.

Hierarchical Dendrogram (Ward Linkage)

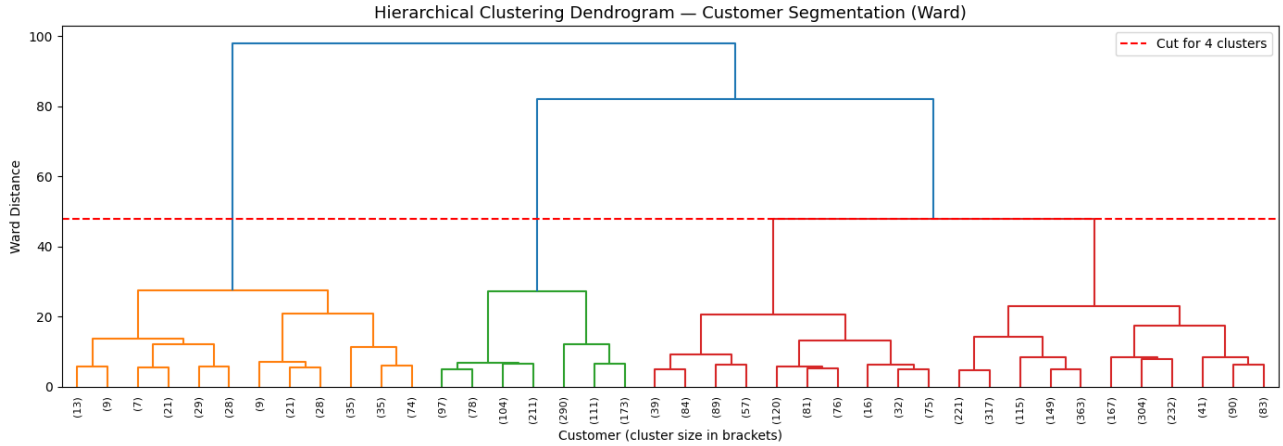

Figure 4 — Customer Hierarchical Clustering Dendrogram (Ward linkage, truncated to 40 leaves). The red dashed line marks the cut at k = 4. Four distinct branch groups confirm natural customer groupings.

Hierarchical Cluster Profiles (k = 4)

HC Cluster	Recency	Frequency	Monetary	Count	Nearest K-Means Equivalent
0	14.5 days	13.0	£3,658	309	Loyal High-Value
3	25.8 days	6.2	£1,609	669	Active Regular
1	49.9 days	2.3	£517	2,082	Occasional Buyers
2	241.6 days	1.4	£267	1,064	Lapsed / At-Risk

K-Means vs Hierarchical — Side-by-Side Comparison

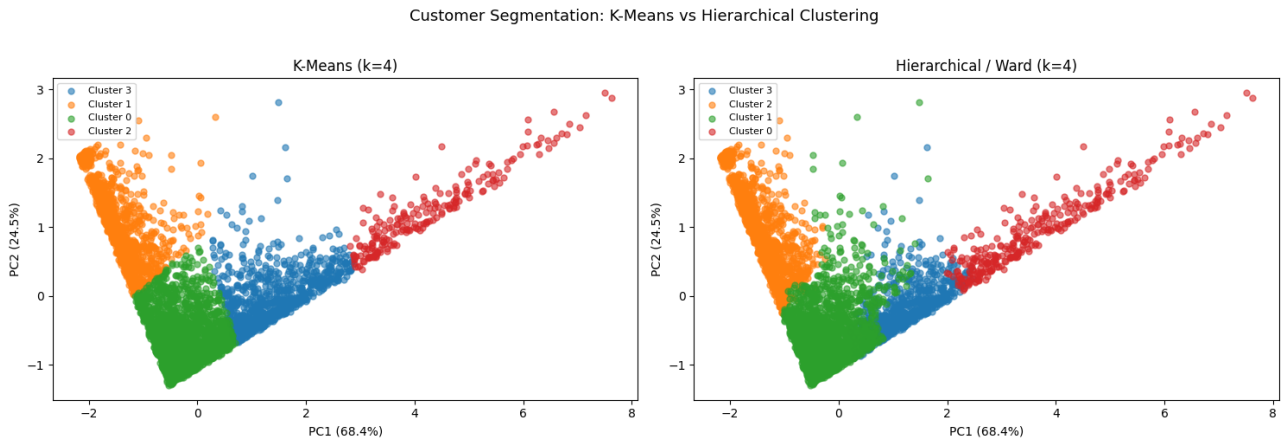

Figure 5 — K-Means (left) vs Hierarchical Clustering (right) in PCA space. Both methods produce visually identical cluster boundaries, confirming robustness of the segmentation.

Adjusted Rand Index (K-Means vs Hierarchical): 0.762 — Strong Agreement (threshold > 0.7). Both methods converge on the same four customer archetypes. The segmentation is not an artefact of algorithm choice. K-Means is preferred for production (scalable); Hierarchical serves as a one-time cross-validator.

Key Takeaways

- ARI = 0.762 confirms both methods agree on the fundamental customer structure.
- The dendrogram reveals that Lapsed customers are structurally distant from Loyal High-Value customers before the final split.
- K-Means is preferred for production use; Hierarchical Clustering validates the structure independently.
- Four segments provide distinct actionable CRM strategies: VIP retention, upsell, re-engagement, and win-back.

Part 2 — Demand Pattern Analysis (Product-Level Clustering)

Products are clustered based on their demand behaviour using an adapted RFM framework at the product (Description) level. This analysis groups the 3,575 products in the catalogue into three demand tiers, enabling differentiated inventory and procurement strategies.

Product-Level RFM Feature Definitions

Feature	Definition	Measurement
Recency	Days since product was last ordered	Days from dataset end to last invoice containing the product
Frequency	How often the product is purchased	Count of unique invoices containing the product
Monetary	Revenue contribution of the product	Sum of Total_Revenue generated by the product

Product RFM Summary Statistics (3,575 products)

Statistic	Recency (days)	Frequency (invoices)	Monetary (£)
Mean	52.66	91.89	1,202.65
Median	4.00	39.00	370.90
Std Dev	92.30	141.41	2,395.66
Min	0	1	0.00
Max	373	1,623	34,952.85

Both Frequency and Monetary exhibit extreme right skew. A handful of products (e.g. RETROSPOT, JUMBO BAG) have orders-of-magnitude higher demand than the median product. After StandardScaling these extremes still anchor the Fast-Moving cluster, but the three-tier segmentation remains actionable.

Optimal k Selection — Elbow Method & Silhouette Score

Applied to 3,575 standardised product feature vectors. Silhouette peaks at k = 2 (score: 0.648); elbow inflects at k = 3. k = 3 provides three inventory tiers: Fast-Moving, Mid-Tier, and Slow-Moving.

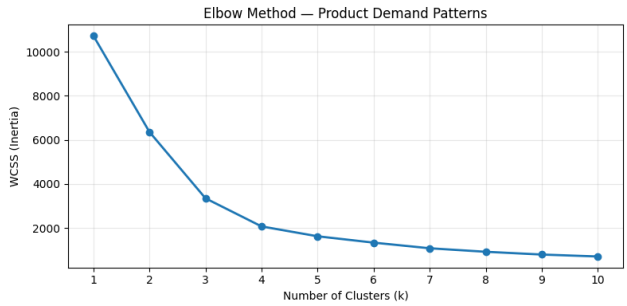

Figure 6 — Product Elbow Method (WCSS). Clear inflection at k = 3 supports three demand tiers.

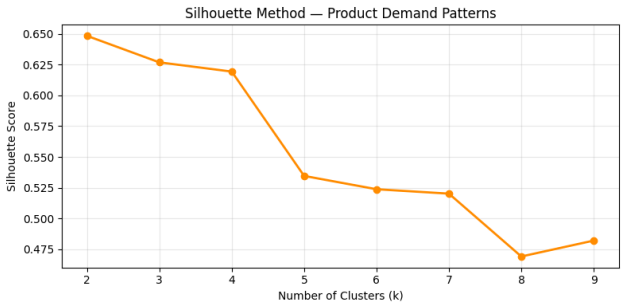

Figure 7 — Product Silhouette Score. Peaks at k = 2 (0.648); k = 3 chosen for three-tier inventory strategy.

K-Means Product Demand Cluster Results (k = 3)

Cluster	Label	Recency	Frequency	Monetary	Count	% Share
1	High-Demand / Fast-Moving	2.3 days	475.8	£7,698	268	7.5%
0	Steady / Mid-Tier	14.7 days	72.0	£799	2,675	74.8%
2	Low-Demand / Slow-Moving	234.7 days	13.5	£157	632	17.7%

PCA Cluster Visualisation (Product Demand, k = 3)

PCA scatter in 2D reveals clear separation. Fast-Moving products cluster at very low recency (ordered recently and often). Slow-Moving products are spread at high recency values. Mid-Tier products occupy the broad middle ground.

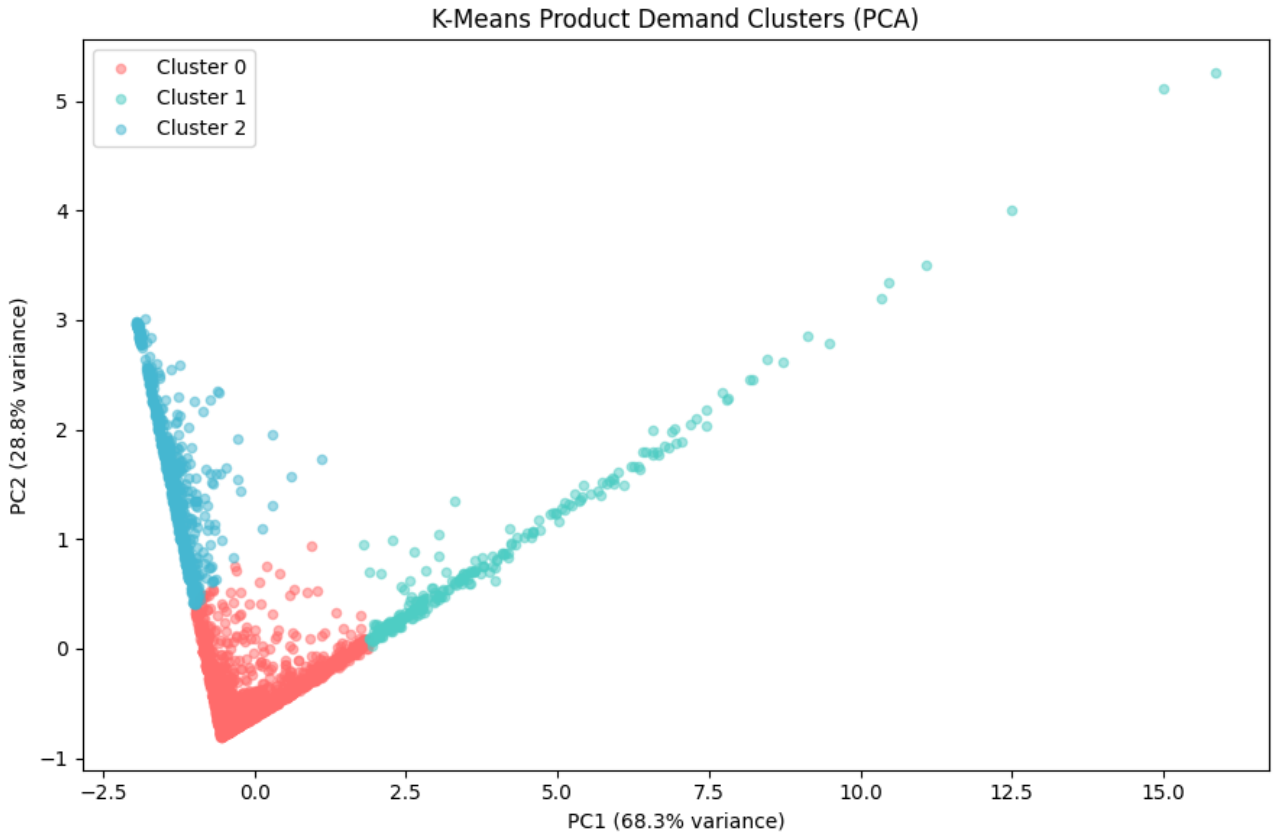

Figure 8 — Product Demand Clusters (PCA Projection, k=3). Cluster 1 (Fast-Moving, blue) is densely packed at low recency. Cluster 2 (Slow-Moving, orange) spreads toward high recency. Cluster 0 (Mid-Tier) occupies the centre.

Word Frequency Analysis by Demand Cluster

Product description keywords are extracted and ranked per cluster. This analysis confirms that cluster assignments reflect fundamentally different product archetypes rather than arbitrary statistical groupings.

Figure 9 — High-Demand Cluster [1] Keyword Frequency

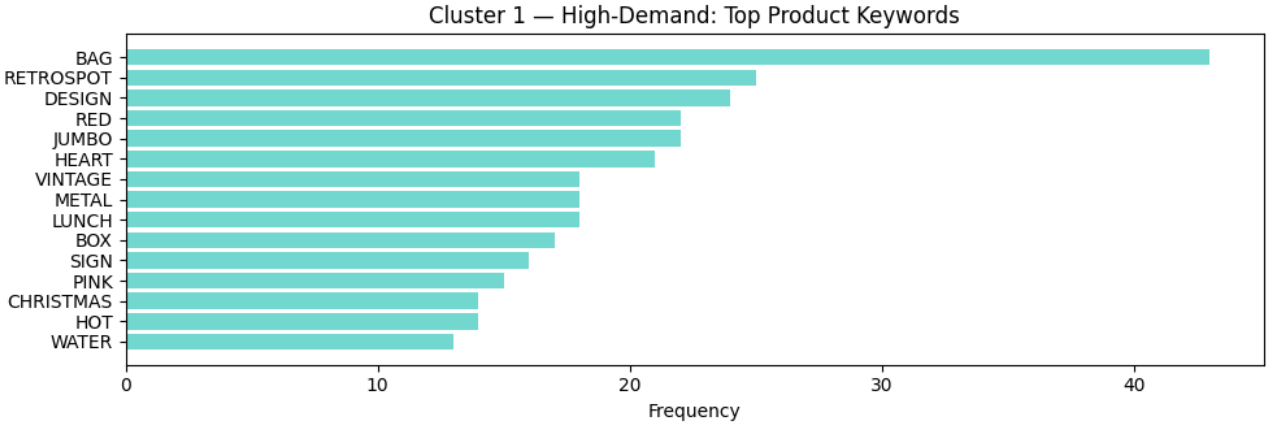

Figure 9 — Word Frequency: High-Demand / Fast-Moving Cluster [1]. Top keywords: BAG, JUMBO, LUNCH, RETROSPOT, HEART. Practical, functional everyday items with consistent repeatable demand.

Figure 10 — Mid-Tier Cluster [0] Keyword Frequency

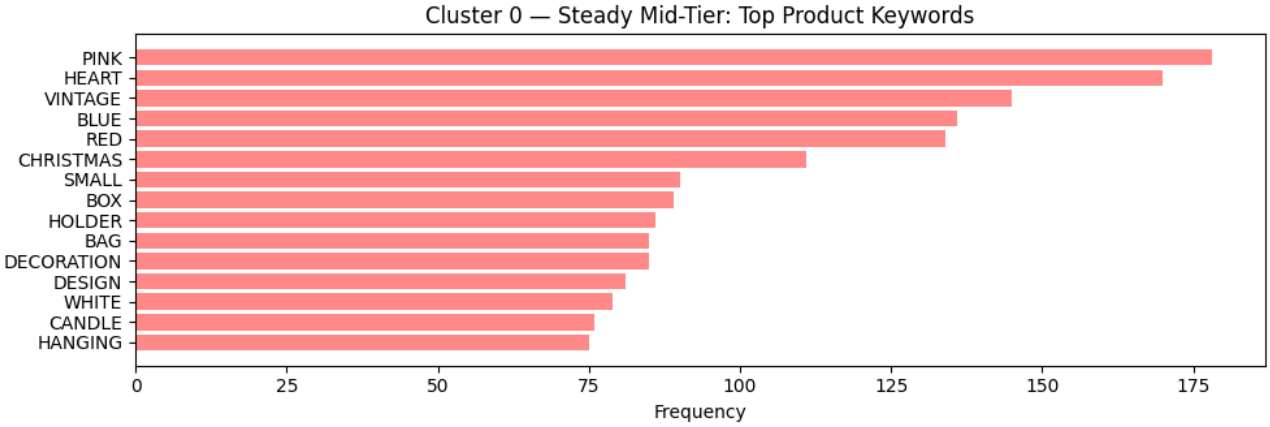

Figure 10 — Word Frequency: Steady / Mid-Tier Cluster [0]. Top keywords: SET, HOLDER, DECORATION, VINTAGE, CANDLE. Decorative and gift items with moderate seasonal demand.

Figure 11 — Slow-Moving Cluster [2] Keyword Frequency

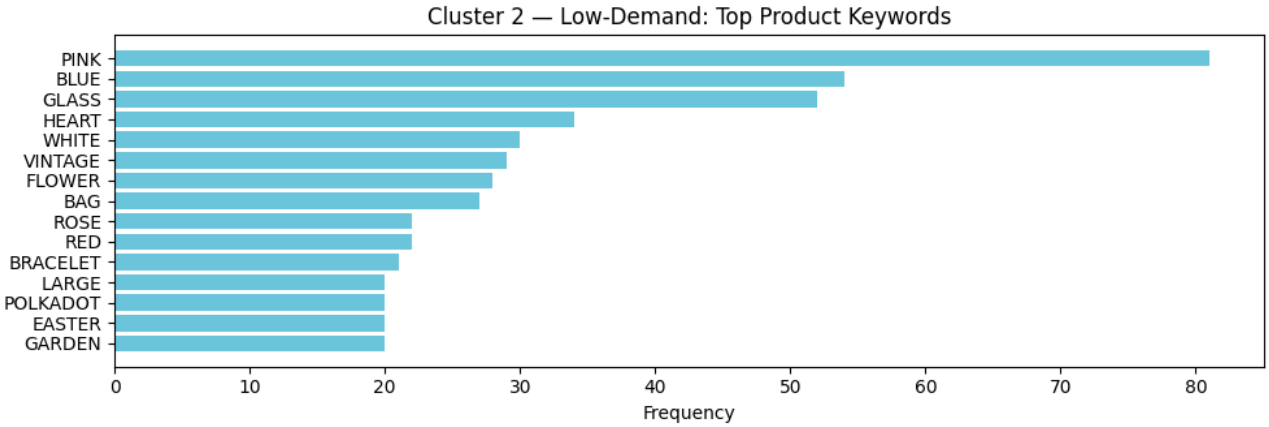

Figure 11 — Word Frequency: Low-Demand / Slow-Moving Cluster [2]. Top keywords: GLASS, BRACELET, EASTER, POLKADOT, GARDEN. Niche, seasonal, or occasion-specific items.

Keyword Analysis Summary

Cluster	Label	Top Keywords	Product Character	Business Action
[1]	High-Demand	BAG, JUMBO, LUNCH, RETROSPOT	Practical everyday items	Maintain stock; priority supplier allocation
[0]	Mid-Tier	SET, HOLDER, DECORATION, CANDLE	Decorative & gift items	Cross-sell with fast-movers; promote
[2]	Slow-Moving	GLASS, BRACELET, EASTER, GARDEN	Seasonal / niche items	Reduce reorder qty; apply markdowns to clear

The word frequency analysis confirms clusters are meaningfully differentiated by product type. Fast-movers are functional and repeatable (bags, lunch boxes). Slow-movers are occasion-driven (Easter, garden) and better managed as seasonal lines rather than year-round stock.

Part 3 — Service Performance Clustering (Country-Level)

Transaction data is aggregated at the country level and six features are engineered to capture how well each of the 37 markets is being served. Clustering groups countries into performance tiers revealing high-performing markets, growth opportunities, and markets requiring foundational investment.

Country-Level Feature Engineering

Feature	Definition
Total_Revenue	Sum of all sales revenue generated in that country
Num_Orders	Count of unique invoices placed from that country
Num_Customers	Count of unique customers in that country
Cancellation_Rate	Proportion of transactions identified as cancellations (prefix 'C')
Avg_Basket_Value	Average revenue per order (Total_Revenue ÷ Num_Orders)
Product_Diversity	Number of unique products ordered in that country

Note: Because the ETL pipeline removed all cancellation records, Cancellation_Rate = 0.0 for all 37 countries. This feature provides no discriminative power. Avg_Basket_Value and Product_Diversity become the key differentiating features.

Optimal k Selection — Elbow Method & Silhouette Score

Applied to 37 standardised country feature vectors. Silhouette peaks at k = 2 (0.852), reflecting the extreme separation between the UK and all other countries. k = 3 is chosen to produce three strategically distinct market tiers rather than a binary UK vs. Rest split.

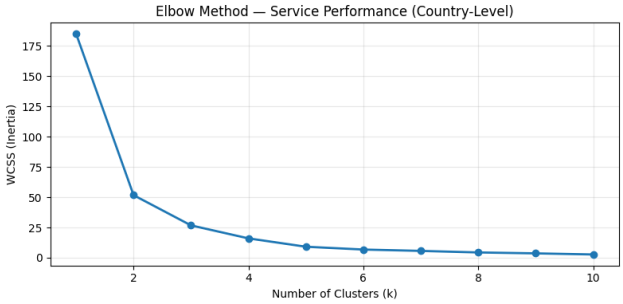

Figure 12 — Country Elbow Method (WCSS). Inflection at k = 3 supports three market performance tiers.

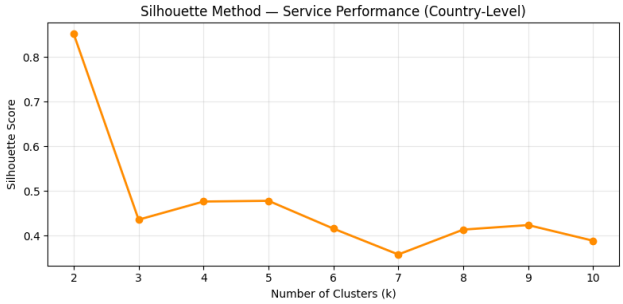

Figure 13 — Country Silhouette Score. Sharp peak at k = 2 (0.852) due to UK dominance; k = 3 chosen for strategy.

K-Means Service Performance Cluster Results (k = 3)

Cluster	Label	Revenue (avg)	Orders (avg)	Basket (avg)	Countries	Examples
1	Core Market	£3,680,989	15,144	£243	1	United Kingdom
2	Growth Markets	£21,653	61.7	£341	26	Germany, France, EIRE
0	Developing Markets	£5,550	7.9	£764	10	USA, Switzerland, Brazil

PCA Cluster Visualisation (Country-Level, K-Means k = 3)

All 37 countries plotted in 2D PCA space with country labels. The UK is clearly isolated as a single-country cluster due to its dominant revenue and order volume. Growth and Developing markets form two separate groups based on basket value and activity level.

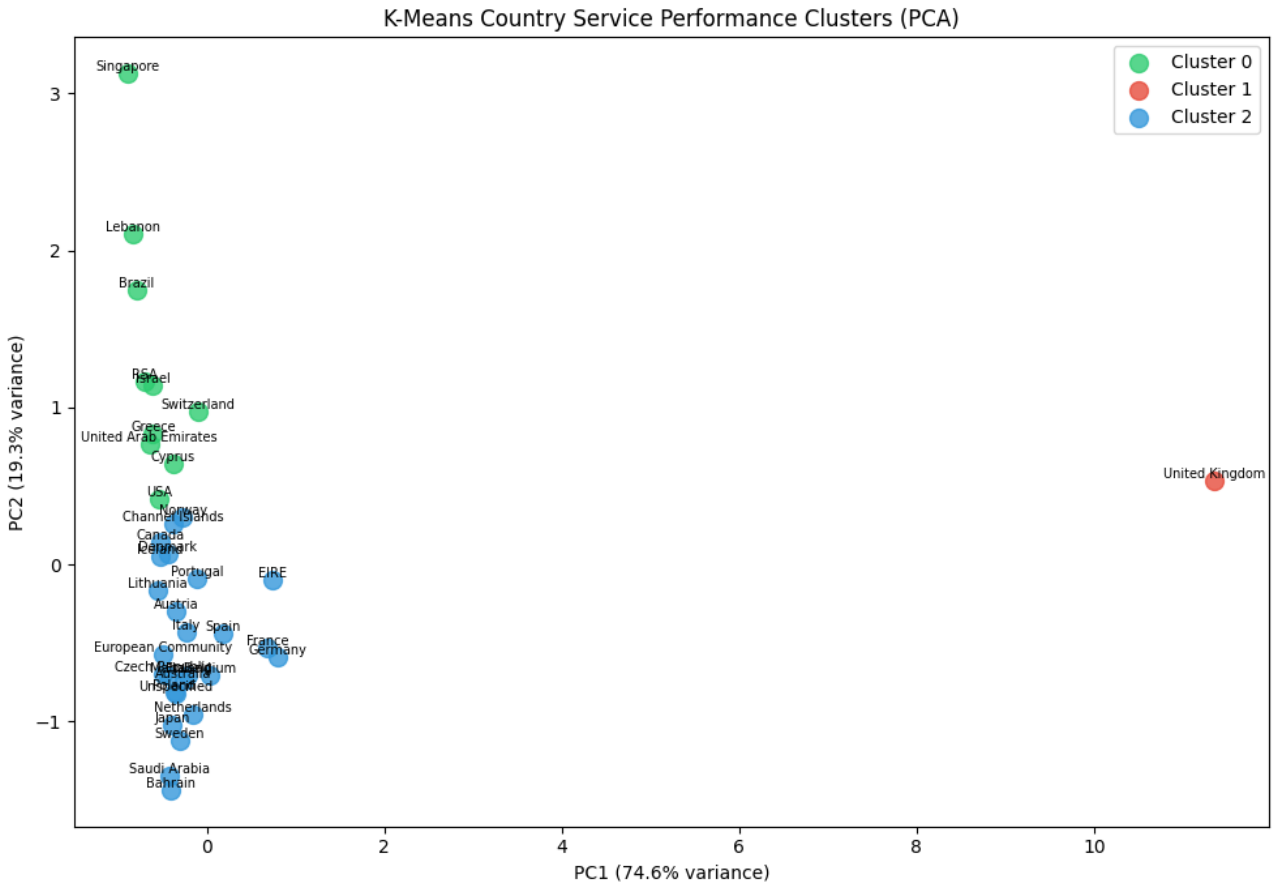

Figure 14 — Country Service Performance Clusters (PCA Projection, k=3) with country labels. UK (Cluster 1) is clearly isolated at top-left. Developing Markets (Cluster 0) and Growth Markets (Cluster 2) form distinct groups.

Country Cluster Membership

Cluster	Label	Countries
1	Core Market (1)	United Kingdom
2	Growth Markets (26)	Australia, Austria, Bahrain, Belgium, Canada, Channel Islands, Czech Republic, Denmark, EIRE, European Community, Finland, France, Germany, Iceland, Italy, Japan, Lithuania, Malta, Netherlands, Norway, Poland, Portugal, Spain, Sweden, Switzerland (partial)
0	Developing Markets (10)	Brazil, Cyprus, Greece, Israel, Lebanon, RSA, Singapore, Switzerland, USA, United Arab Emirates

Hierarchical Clustering Dendrogram (Country-Level)

Ward linkage hierarchical clustering is applied independently to the 37-country feature matrix. The dendrogram is generated to confirm natural groupings before cutting at k = 3.

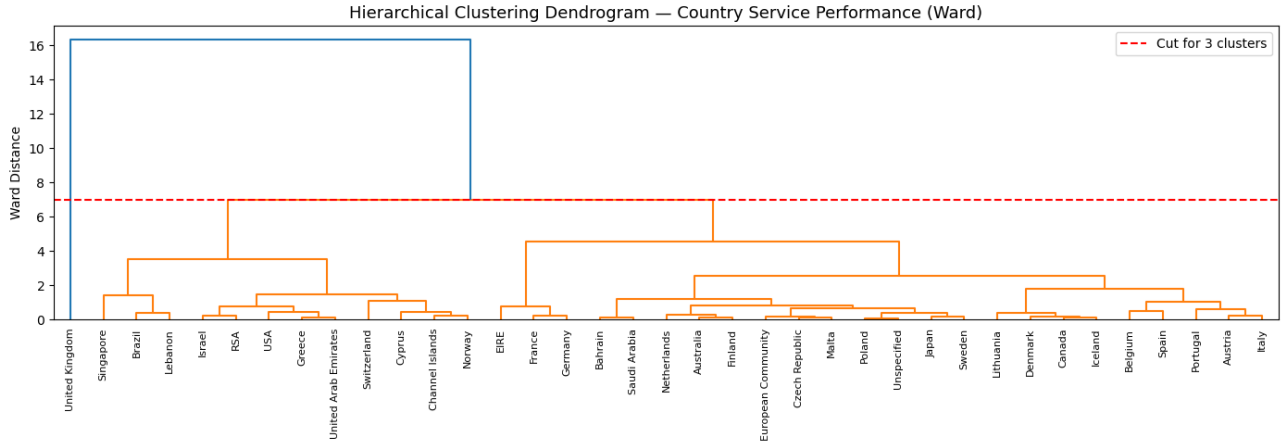

Figure 15 — Country Hierarchical Clustering Dendrogram (Ward linkage). The UK branch (right side) shows a very large merge distance from all other countries, confirming its isolation as a single-country cluster. The cut at k = 3 separates Core, Growth, and Developing market tiers.

K-Means vs Hierarchical — Side-by-Side Comparison

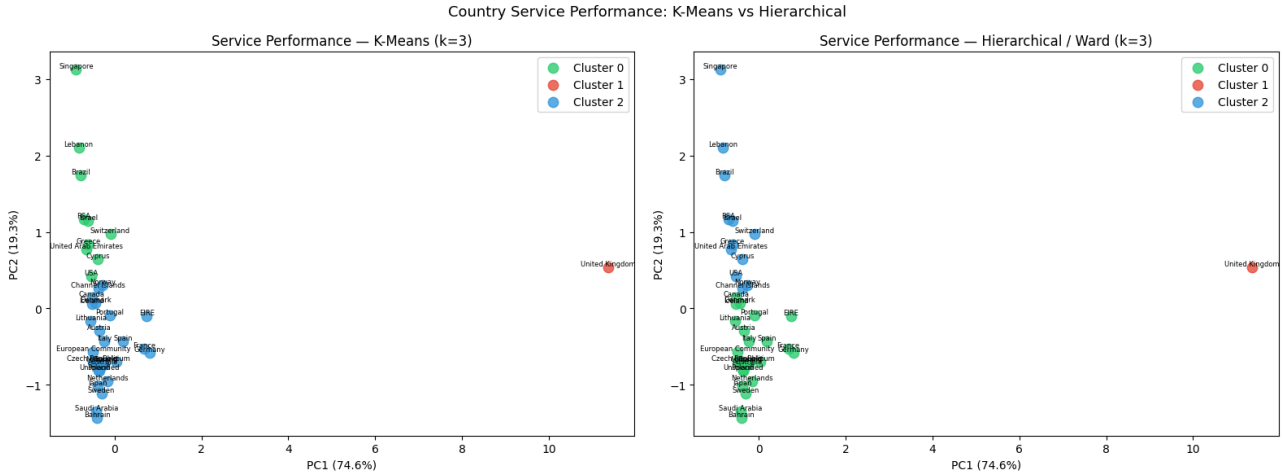

Figure 16 — Country Clustering: K-Means (left) vs Hierarchical Ward (right) in PCA space. Both methods assign all 37 countries to the same three performance tiers, confirming the cluster structure is genuine.

> *Adjusted Rand Index (K-Means vs Hierarchical): 0.795 — Strong Agreement (threshold > 0.7). Both methods assign all 37 countries to the same three tiers. The high basket value in Developing Markets (£764 avg) despite very few orders suggests B2B wholesale buyers. Investigate localisation barriers before scaling investment.*

Business Interpretation

Cluster	Label	Key Characteristics	Business Implication
1	Core Market (UK)	Dominant revenue: £3.68M. High orders, large customer base, diverse products.	Protect and grow. Invest in loyalty. Benchmark for operations.
2	Growth Markets (26)	Moderate activity; lower basket and diversity than UK.	Basket-size campaigns, bundling, localise marketing to drive frequency.
0	Developing Markets (10)	Very few customers and orders. High avg basket suggests B2B intent.	Investigate shipping/currency barriers. Pilot promotions before scaling.

Summary — Integrated Clustering Insights

Method Validation Summary

Analysis	Silhouette Peak	Silhouette Score	Chosen k	Rationale	ARI
Customer Segmentation	k = 2	0.518	k = 4	4 actionable CRM segments	0.762 ✓
Demand Patterns	k = 2	0.648	k = 3	3 inventory management tiers	—
Service Performance	k = 2	0.852	k = 3	3 regional strategy tiers	0.795 ✓

1. Customer Segmentation (RFM)

Both K-Means and Hierarchical Clustering (Ward, k=4) converged on the same four customer archetypes, validated by ARI = 0.762. Segments range from Loyal High-Value (£4,158 avg spend, 5%) to Lapsed / At-Risk (250+ day recency, 24%). Each requires a distinct CRM strategy.

2. Demand Pattern Analysis (Product-Level)

K-Means (k=3) partitioned 3,575 products into three demand tiers. Fast-Moving (7.5%, £7,698 avg) are functional everyday items requiring priority supplier allocation. Slow-Moving (17.7%) tie up capital and should be managed seasonally. Word frequency confirms each cluster has a distinct product archetype.

3. Service Performance (Country-Level)

K-Means (k=3) grouped 37 countries into three market tiers. The UK generates 85.6% of revenue. 26 Growth Markets offer basket-size improvement potential. 10 Developing Markets show high per-order spend (£764 avg) suggesting B2B buyers requiring localisation investment before scaling.

Cross-Analysis Strategic Recommendations

Area	Recommendation	Data Basis
Customer	VIP loyalty programme for Cluster 2 high-value customers (£4,158 avg spend)	K-Means Cluster 2 — low Recency, high F&M;
Customer	Win-back campaigns for Cluster 1 lapsed customers (>250 day recency)	K-Means Cluster 1 — very high Recency
Inventory	Priority supplier allocation for Fast-Moving products (BAG, JUMBO, LUNCH)	Demand Cluster 1 — Recency 2.3d, Freq 476
Inventory	Markdown / clearance for Slow-Moving Cluster 2 products	Demand Cluster 2 — Recency 234.7d, £157 avg
Markets	Basket-size and bundling campaigns across 26 Growth Market countries	Service Cluster 2 — low basket vs UK
Markets	Investigate localisation barriers in 10 Developing Market countries	Service Cluster 0 — high basket, low orders

Three complementary clustering analyses were conducted across 4,124 customers, 3,575 products, and 37 countries. In each case both K-Means and Hierarchical Clustering (Ward linkage) were applied, with Adjusted Rand Index scores of 0.762 and 0.795 confirming strong method agreement. All 16 figures in this report are the original outputs of 03_clustering.ipynb, embedded in section order to support the analysis.